Pengyu Wang

Full-Stack Developer

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PROFESSIONAL SUMMARY

Full-Stack Developer with an MSc in Computational Science, experienced in building mobile and web applications using Java, JavaScript, Python, and SQL. Also have experience in Python-based AI abilities. Strong foundation in system design, data structures, and scalable application development. Passionate about developing efficient, user-focused software solutions.

TECHNICAL SKILLS

Languages

Java • JavaScript • Python • Kotlin • C++ • Swift • SQL

Frontend

HTML • CSS • JavaScript • Android XML • iOS Storyboard

Backend & APIs

Node.js • RESTful APIs • Object-Oriented Programming • System Design

Databases

MySQL • SQLite • HyperSQL (HSQLDB)

Testing

JUnit4/5 • XCTest • Unit Testing • Integration Testing

Tools & Technologies

Git • Android Studio • Xcode • Gradle • Linux

AI / Data (Optional but valuable)

Scikit-learn • Pandas • NumPy • Matplotlib

PROJECTS

CRM SYSTEM (FULL STACK) *Toronto, ON, 2026*

- Designed and developed a lightweight **cross-platform CRM application (Web, Android, and iOS)** to manage automotive sales leads, interaction tracking, and follow-up workflows for 100+ customers, enabling **structured customer lifecycle management**, priority-based task scheduling, and scalable workflow automation.
- Implemented a lead prioritization algorithm combining stage weighting, engagement scoring, and time-decay urgency modeling.
- Built a **Priority Queue–based ranking engine** enabling real-time lead prioritization and eliminating repeated dataset sorting.
- Integrated a **MySQL persistence layer** to store lead profiles, interaction history, and follow-up schedules, enabling **structured data management**, and **scalable retrieval** for **priority ranking** and **reminder automation**.
- Developed automated follow-up reminders and overdue detection features based on lead priority scoring and timeline tracking logic, improving **deal close rates from 62% to 88%** through systematic follow-up automation and supporting more consistent, timely customer engagement workflows.

IMMIGRATION DRAW PREDICTION MODEL (PYTHON MACHINE LEARNING) *Toronto, ON, 2025*

- Developed a **machine learning prediction model** for immigration draw score forecasting using public government datasets.
- Implemented predictive models using **Python, Scikit-learn, and Pandas**.
- Built automated data preprocessing pipelines, reducing data processing time by 30%.
- Designed data visualization dashboards using **Matplotlib and Seaborn** to improve result interpretation and decision-making.

MULTIPLAYER MATCHMAKING SYSTEM (RANKING SIMULATION ENGINE) *Sudbury, ON, 2022-2024*

- Designed a **fairness-optimized matchmaking algorithm** for large-scale multiplayer environments, modeling **player skill distributions** to improve match balance and responsiveness.
- Built modular simulation pipelines to evaluate matchmaking performance **across 100+ experimental scenarios**.
- Implemented adaptive ranking strategies balancing latency, fairness, and match quality tradeoffs.
- Improved matchmaking stability metrics by **~18% over baseline approaches**.

ONLINE FOOD ORDERING PLATFORM *Winnipeg, MB, 2020*

- Developed a user-friendly **online food ordering platform** using JavaScript, HTML, and CSS, enabling customers to browse menus and place orders through an interactive web interface.
- Implemented responsive UI components and client-side logic to improve navigation efficiency and order selection accuracy.
- Deployed the application on a **Node.js runtime environment** to support backend communication and system stability.
- Conducted usability testing with 10 users, improving order accuracy and user interaction efficiency.

EDUCATION

Laurentian University- **MSc in Computational Science** *Sudbury, ON, 2022 - 2024*

- Engineering and Computer Science Graduate Fellowship
- Ontario Scholarship 2024

University of Manitoba- **BSc. of Computer Science, Minor in Math** *Winnipeg, MB, 2019 – 2022*

- Relevant Coursework: **Software Engineering**; Data Structures & Algorithms; **Databases**; **Object-Oriented Programming**

EXPERIENCE

Graduate Teaching Assistant & Technician

Laurentian University *Sudbury, ON, 2023 - 2024*

- Assisted in delivering a Python-based introductory programming course, supporting instruction on control flow, functions, object-oriented programming, and basic data structures.
- Held lab sessions and debugging workshops for 70+ students, guiding them through hands-on implementation of file I/O, data processing, and modular program design. Helped students troubleshoot runtime errors, logical bugs, and performance issues, reinforcing systematic debugging strategies. Reviewed and provided feedback on student code, emphasizing clean architecture, readability, and algorithmic efficiency.
- Collaborated with course instructors to prepare lab materials, assignment specifications, and grading rubrics.